ATOMIC STRUCTURE AND CHEMICAL BONDING

📖 1. Core Definitions & Key Terminology 📝

- **Atom (Modern Scientific Definition):** The smallest particle of an element that exhibits all the properties of that element. It may or may not exist independently but takes part in every chemical reaction. In other words, an atom is the smallest possible unit of an element. ⚛️
- **Element:** A pure substance that can neither be formed from, nor decomposed into simple substances by ordinary physical or chemical methods (e.g., Carbon). 🧱
- **Electron:** A subatomic particle having a unit negative charge and a mass equal to $1/1837$ of a hydrogen atom. It is denoted by the symbol ${}_{-1}e^{0}$, where the superscript 0 represents mass and the subscript -1 represents its electrical charge. ⚡
- **Proton:** A positively charged subatomic particle with a unit positive charge $(+1)$ and a mass equal to that of a hydrogen atom $(1$ a.m.u.$)$. ➕
- **Static (or Frictional) Electricity:** The electrification of objects by rubbing them with each other. ⚡
- **Radioactivity:** The processes of radioactive decay and high-energy nuclear reactions that can transform one elementary substance into another.
    - *Footnote Definition:* The phenomenon due to which certain elements spontaneously emit highly penetrating rays called radiation. ☢️
- **Cathode Rays:** Rays that originate from the negative plate (cathode) and travel towards the anode in a discharge tube when high voltage is applied at very low gas pressures. 🗡️
- **Anode Rays (Canal / Positive Rays):** Rays travelling in a direction opposite to cathode rays (from anode to cathode) through a perforated cathode; they consist of positively charged particles (protons). 🌞

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **The Ancient Indian Concept of Matter:**
    - Maharshi Kanada (6th Century B.C., India) proposed that matter consists of indestructible, minute particles called *paramanus*. 🧘
    - *Word Breakdown: Param* means "ultimate," and *anu* means "particle" (now called atoms). 🧩
    - A *paramanu* does not exist in a free state; rather, it combines with other *paramanus* to form a bigger particle called *anu* (now known as a molecule). 🔗
    - Different types of *paramanus* exhibit specific properties. ✨
- **The Ancient Greek Concept:** Democritus (460 B.C. – 370 B.C.) called the *paramanu* an "atom", derived from the Greek word *atomos*, meaning indivisible. 🏛️
- **Why is Carbon considered an element?** It cannot be split up into two or more simpler substances by ordinary methods like heating, breaking, or passing electricity. 💎
- **The Zinc Grinding Example (Explaining Atoms):** If you take a small piece of zinc and grind it, the smaller pieces still show the properties of zinc. If you grind them further into very fine particles, they still exhibit zinc's properties. A stage comes when the particles cannot be further subdivided while still exhibiting those properties; these indivisible particles are the atoms of zinc. 🪙
- **Why do gases suddenly conduct electricity in a discharge tube?** Gases are ordinarily poor conductors. However, when a massive voltage (10,000 volts) is applied at a very low pressure (0.01 mm Hg), the gases become good conductors, and electricity begins to flow from the cathode to the anode in the form of rays. 🌩️
- **Why do cathode rays prove electrons are negative?** They are inflected towards the positive field and deflected from the negative field in an electric field. 🧲
- **How did Thomson conclude that electrons are an integral part of all atoms?** He found that electrons from all sources are alike, and their mass

and charge remain perfectly constant regardless of the material of the cathode or the nature of the gas enclosed in the discharge tube. 🍼

- **Why must atoms contain positively charged particles (protons)?** Atoms are found to be electrically neutral. Therefore, they must contain positively charged particles such that the total negative charge of the electrons is perfectly balanced by the total positive charge. ⚖️
- **How are protons actually produced in the discharge tube?** When a high-voltage current is passed through a gas in the tube, the electrical energy breaks up the atoms of the gas into negatively charged particles (electrons) and positively charged particles (protons). ⚡
- **Naming the Rays:**
    - *Canal rays:* Named by Goldstein because they passed through holes or "canals" in a perforated cathode. 🕳️
    - *Positive rays:* Named because they consist of positively charged particles. ➕
    - *Anode rays:* Named because they are directed away from the anode. ➡️
- ✍️ *Handwritten Margin Note:* "Half-Yearly" (written at the top of the chapter title page). 📝
- ✍️ *Handwritten Margin Note:* "e/m" (written in the properties of electrons section, reinforcing the charge-to-mass ratio concept). 📝
- 🧩 *Page-Shift Stitch:* Properties of Cathode Rays: "5. They penetrate through matter." (Bottom of page 78) seamlessly connects to "6. They cause ionization of the gas through which they pass." (Top of page 79). 🔚

📌 3. Rules, Principles & Characteristics ⚙️

- **Main Postulates of Dalton's Atomic Theory (1808):**
    - Matter consists of very small and indivisible particles called atoms. 🔬
    - Atoms can neither be created nor be destroyed. 🛡️

    - Atoms of an element are alike in all respects but they differ from the atoms of other elements. 👥
    - Atoms of an element combine in small numbers to form molecules. 🔗
    - Atoms of one element combine with atoms of another element in a simple ratio to form molecules of compounds. 🤝
    - *Correction Note:* Latest research has proved that most of the postulates of Dalton's atomic theory are incorrect, except that atoms are the smallest units that take part in chemical reactions. ❌✅
- **Properties of Cathode Rays:**
    - Travel from cathode to anode in straight lines. 📏
    - Cause a greenish yellow fluorescence on a soda-glass screen placed in the tube. 🟢
    - Affected by an electric field (deflected towards the positive plate). 🧲
    - Produce X-rays when made to fall upon hard metallic targets like tungsten. ☢️
    - Penetrate through matter. 🧱
    - Cause ionization of the gas through which they pass. ⚡
    - Produce the shadow of an opaque object placed in their path and make a light paddle wheel rotate (proving they have mass/momentum). 🎡
    - The charge-to-mass ratio (e/m) remains the same irrespective of the gas taken or the metal forming the cathode. ⚖️
- **Properties of Electrons:**
    - Electrons from all sources are alike, having identical mass. 👯
    - They are a constituent part of all atoms. 🧩

    - They carry a unit negative charge and have an extremely small mass. ➖
    - The charge and mass of an electron always remain constant. 🔒
- **Properties of Anode (Canal) Rays:**
    - Travel in a straight line. 📏
    - Consist of minute material particles and hence produce mechanical effects. ⚙️
    - Made up of positively charged particles. ➕
    - Deflected by electric and magnetic fields in a direction opposite to that of cathode rays (confirming positive charge). 🧲
    - Produce fluorescence on a zinc sulphide screen. ✨
    - Their e/m (charge-to-mass) ratio differs from gas to gas. It is much less than that of an electron and is at its maximum when hydrogen is taken in the discharge tube. 📈
- **Properties of Protons:**
    - A proton possesses a unit positive charge $(+1)$. ➕
    - Its mass is the same as that of a hydrogen atom. ⚖️

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Timeline of Key Discoveries:**
    - **1808:** John Dalton gives the first scientific theory about the structure of matter. 🗓️
    - **1833:** Faraday shows that the flow of electricity is due to the flow of charged particles. 🗓️
    - **1874:** G.J. Stoney suggests the name 'electron' (meaning 'atom of negative electricity') for this electrical particle. 🗓️
    - **Late 19th Century:** William Crookes discovers cathode rays using a high-voltage discharge tube. 🗓️

- **Late 19th Century:** J.J. Thomson studies the characteristics of cathode rays and discovers the electron. 🗓️
- **1886 (Implied):** E. Goldstein discovers canal rays (protons) using a perforated cathode. 🗓️
- **Later:** James Chadwick discovers neutrons. 🗓️

- **Discharge Tube Experimental Conditions:**
  - *Voltage applied:* $10{,}000$ volts (from an induction coil). ⚡
  - *Pressure:* Very low, 0.01 mm Hg (created by a vacuum pump). 📉

- **Subatomic Particles Master Data Table:**

Particle	Symbol	Mass (Relative & Exact)	Charge (Magnitude)	Key Features
Electron	${}_{-1}e^{0}$	$1/1837$ of a Hydrogen atom $\approx 9.108 \times 10^{-31}$ kg	Unit Negative (-1) -1.602×10^{-19} coulombs	Radius $< 1 \times 10^{-15}$ m; Constant e/m ratio of 1.76×10^{11} C/kg.
Proton	p^{+}	1 a.m.u. $\approx 1837 \times$ mass of electron	Unit Positive $(+1)$ $+1.602 \times 10^{-19}$ coulombs	e/m ratio varies based on the gas in the tube; max with Hydrogen.

Particle	Symbol	Mass (Relative & Exact)	Charge (Magnitude)	Key Features
		$\approx 1.672 \times 10^{-24}$ g		

📖 1. Core Definitions & Key Terminology 📝

- **Proton:** A subatomic particle having mass 1 amu (i.e., equal to a hydrogen atom) and possessing a unit positive charge. It is denoted as $_{+1}p^{1}$, where superscript 1 represents mass (1 amu) and subscript +1 represents one unit positive charge. ➕
- **Nucleus:** The positively charged mass at the centre of the atom where the entire mass of the atom is concentrated; it is the densest part of the atom. 🎯
- **Energy Level (Shell or Orbit):** Certain fixed circular orbits around the nucleus where revolving electrons are confined; each is associated with a fixed amount of energy. 🪐
- **Alpha (α) Particle:** A doubly charged helium ion (He^{2+}), containing two protons and two neutrons. It is formed by removing 2 electrons from a helium atom.
    - *Footnote Definition:* An alpha particle consists of two protons and two neutrons. It is the same as a Helium atom that has lost both its electrons. ⚛️
- **Neutron:** A sub-atomic particle or fundamental particle of an atom with no charge and mass almost equal to the mass of the proton (i.e., hydrogen atom). It is denoted by $_{0}n^{1}$. The superscript 1 represents its mass and subscript 0 represents its electrical charge. ⚪

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why do anode rays originate differently than cathode rays?** Anode rays never originate from the anode itself; they consist of the positive residues left when electrons are knocked out of the atoms of the gases

enclosed in the discharge tube. In contrast, cathode rays originate from the metals or electrodes which constitute the cathode. ⚡

- **Why did Rutherford choose gold for his experiment?** In 1911, Lord Rutherford selected a gold foil because he wanted a very thin layer (one millionth of a centimetre), and gold is the most malleable metal. 🥇
- **How did Rutherford prove the existence of an empty space in the atom?** He observed that most of the alpha particles passed straight through the metal foil without deviating from their path. ➡️
- **How did Rutherford prove the nucleus contains protons (positive charge)?** Very few α-particles were deflected by large angles or bounced back. Since α-particles are positively charged, their deflection proved they were repelled by a concentrated, positively charged mass at the centre (the nucleus). 🧲
- **How did Rutherford prove the size of the nucleus is tiny?** The size of the nucleus is very small compared to the size of the atom, proved by the fact that the vast majority of the α-particles passed straight through the gold foil without hitting anything. 👏
- **Why is an atom electrically neutral?** The total positive charge of the nucleus (protons) is perfectly balanced by the total negative charge of all the revolving electrons. ⚖️
- **Why doesn't the electron fall into the nucleus due to attraction?** The force of attraction between the negatively charged electrons and the positively charged nucleus is counterbalanced by the centrifugal force acquired by the fast-revolving electrons in their circular paths. 💫
- **Why was Rutherford's model fundamentally unstable (Drawback)?** According to classical laws of mechanics and electrodynamics, if an electrically charged particle (electron) is in continuous motion (revolving around the nucleus), it should inevitably radiate energy. Losing energy would cause it to be gradually pulled towards the nucleus in a spiral path, resulting in a collision and the total collapse of the atom. 📉💥
- **Why did the discovery of the neutron become necessary?** Knowing the atom contains electrons (negligible mass) and protons (1 a.m.u.), a helium

atom with 2 protons should logically have a mass of 2 a.m.u. However, its actual atomic mass was approximately 4.0 a.m.u. Therefore, it was proposed that the nucleus must contain another particle with mass equal to a proton but no electric charge to make up the missing mass. ⚖️

- **How did Chadwick discover neutrons (1932)?** He discovered them by bombarding light nuclei like beryllium with alpha (α) particles (i.e., helium nuclei). 💥
- *Note on Heavy vs. Light Metals in Rutherford's Experiment:* Heavy metals such as platinum show the same observation as gold foil. But if light nuclei like lithium are used, fast-moving α-particles may even push the light nucleus aside and may not be deflected back. 🧱

📌 3. Rules, Principles & Characteristics ⚙️

- **J.J. Thomson's 'Plum Pudding' Model (1904):**
  - An atom is considered to be a sphere of uniform positive charge. 🔴
  - Electrons are embedded into this positive sphere (like plums in a pudding). 🫐
  - Total positive charge equals total negative charge (atom is neutral). ⚖️
  - The mass of an atom is considered to be uniformly distributed. 🌐
  - **Limitations:** Failed to address the concept of a central nucleus, couldn't explain how positive and negative charges were held together without collapsing (stability), and gave no insight into electron arrangement or chemical bonding. ❌
- **Rutherford's Atomic Model (Based on Gold Foil Experiment):**
  - The atom contains a large empty space (extra-nuclear space). 🌌
  - There is a positively charged, extremely dense mass at the centre called the nucleus, containing the entire mass of the atom (protons). 🎯

- The size of the nucleus is very small compared to the whole atom. 👊
- Electrons revolve around the nucleus at large distances in circular orbits found in the empty space. 💫
- An atom as a whole is electrically neutral. ⚖️
- Electrons revolve in circular paths (ORBITS); electrostatic attraction is balanced by centrifugal force. 🔁
- *Analogy:* Similar to the solar system (Sun = nucleus, Planets = electrons). 🌞🪐

- **Drawbacks of Rutherford's Atomic Model:**
  - Failed to explain the stability of an atom (radiating energy issue leading to collapse). 💥
  - Failed to explain the distribution of electrons around the nucleus and why they don't fall in. ❓
- **Main Postulates of Bohr's Atomic Model (1913):**
  - Electrons revolve around the nucleus confined to certain fixed orbits called shells or energy levels, each associated with a fixed amount of energy. 🪐
  - While revolving in a particular orbit, an electron neither loses nor gains energy. 🛑
  - An electron jumps to the next orbit by gaining a certain amount of energy, and vice versa. Since each orbit has fixed energy, Bohr called it an *energy level*. ⚡
  - Energy levels are labelled K, L, M, N or $1,2,3,4$, etc. The orbit closest to the nucleus is K shell (least amount of energy). 🔢
- **Properties of Neutrons:**
  - Not deflected by any magnetic or electric field, proving it is electrically neutral. 🧲🚫
  - Its mass is equal to 1.676×10^{-24} g (1 amu). ⚖️

- **Basic Structure of the Atom (Summary):**
    - Electrons (negative) are found outside the nucleus. ➖
    - Protons (positive) are found in the nucleus. ➕
    - Neutrons (neutral) are found in the nucleus and are slightly heavier than protons. ⚪

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Proton Formation Equation:**
    - $H(\text{hydrogen atom}) - e^- \rightarrow H^+(\text{proton})$ ⚡
- **Neutron Discovery Reaction (Chadwick):**
    - Bombarding Beryllium with Alpha particles:
    - ${}_4Be^9 + {}_2He^4(\alpha) \rightarrow {}_6C^{12} + {}_0n^1(\text{neutron})$ 💥
- **Comparison: Dalton's Atomic Theory vs. Modern Atomic Theory**

Dalton's Atomic Theory	**Modern Atomic Theory**
(i) Atoms are indivisible particles.	(i) Atoms are divisible into sub-atomic particles like protons, electrons and neutrons.
(ii) Atoms can neither be created nor destroyed.	(ii) Atoms can be created and destroyed by nuclear fusion and fission. 💥
(iii) The atoms of an element are alike in all respects.	(iii) The atoms of an element may not be alike in all respects, as is seen in the case of isotopes. ⚖️ (Handwritten tick mark)
(iv) Atoms of elements combine in small whole numbers to form molecules.	(iv) In the formation of organic compounds, the numbers can be very high, such as thousands. 🔗 (Handwritten cross mark)

📖 1. Core Definitions & Key Terminology 📝

- **Atomic Number (Z):** An atom of an element has its own characteristic number of protons in its nucleus, which distinguishes it from the atoms of other elements. This characteristic number (number of protons) is called the atomic number of the element. 🔢
    - *Equation Definition:* Atomic number of an element = number of electrons present in its neutral atom = number of protons present in the nucleus of its atom = positive charge in the nucleus of its atom. ⚖️
- **Mass Number (A):** The total number of protons and neutrons present in its nucleus. Mass number is denoted by A. ⚖️
    - *Equation Definition:* Mass Number (A) = no. of neutrons (n) + no. of protons (p).
- **Orbits:** They are the imaginary paths where electrons revolve around the nucleus of the atom. 💫
- **Electronic Configuration:** Distribution of electrons in different shells is called electronic configuration. ⚙️
- **Nucleons:** The protons and neutrons collectively found in the central part or nucleus of the atom. 🎯
- **Nucleus:** The central part of the atom which contains protons and neutrons. 🎯

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why is the atomic number important?** In 1913, Henry Moseley studied how different elements produce different types of X-rays and found that each element has a unique property related to the number of protons in its atom. He discovered that this number, called the atomic number, is what defines an element, not its weight. 💡
- **Why does the atomic number equal the number of electrons?** The atomic number is equal to the number of electrons in the neutral atom of an element because in a neutral atom, no. of protons = no. of electrons. ⚖️

- **How does the atomic number relate to charge?** The number of protons gives the total positive charge present in the nucleus of an atom. Atomic number is unique for atoms of each element. ➕
- **Why is the mass number a whole number?** Mass number, however, is a whole number approximation of the atomic mass calculated in atomic mass units. 🔢
- **What is the relationship between Mass Number and Atomic Mass?** Mass number of an element is nearly the same as its atomic mass numerically. Mass number is merely a number while atomic mass has specific units. The atomic mass is expressed in atomic mass unit (amu) μ. For example, the mass number of sodium is 23 and its atomic mass is 23μ. ⚖️
- **Why does Calcium (Ca) have the configuration $2,8,8,2$ instead of $2,8,10$?** For example, electronic arrangement in Calcium having 20 electrons is $K(2), L(8), M(8), N(2)$. The maximum capacity of M shell is 18 electrons but only 8 electrons are filled in it. This is due to the fact that the elements become stable only after acquiring 8 electrons in the outermost shell. Therefore, the outermost shell of a chemically stable atom can accommodate a maximum of 8 electrons. 🛡️
- **How are electrons filled in heavier elements?** In lighter elements, up to argon (atomic number 18), each inner shell is completely filled before any electron can occupy an outer shell. However, in elements heavier than argon, the situation changes. Although the third shell can accommodate up to 18 electrons, yet the fourth shell begins to be filled after it has only 8 electrons. (Example: Potassium (19 electrons) configuration is $2,8,8,1$). 🀄
- **Example: Structure of Magnesium atom:** Magnesium has atomic number 12 and atomic mass 24. Hence its nucleus has 12 protons and $(24 - 12 = 12)$ neutrons. It is surrounded by 12 electrons that are allotted to different shells as follows: K shell $= 2$ electrons, L shell $= 8$ electrons, M shell $= 2$ electrons. So its electronic configuration is $2,8,2$. 🎗️
- ✍️ *Handwritten Margin Note:* "Learn" (written next to the definition box for Atomic Number). 📝

- ✍️ *Handwritten Margin Note:* "$2n^2$" (written next to the Bohr-Bury rule for maximum electrons). 📝
- ✍️ *Handwritten Margin Note:* "Imp" (Important, written with a tick mark and circle above "Main features of the structure of atoms"). 📝
- 🧩 *Page-Shift Stitch:* "Modern theories about the atom include the existence of over 20 subatomic particles such as positrons, neutrinos, etc." (Footnote seamlessly stitched from the bottom of page 83). 🔚

📌 3. Rules, Principles & Characteristics ⚙️

- **Main features of the structure of atoms:**
    - (i) Atoms of all the elements (except hydrogen) are made up of three fundamental (sub-atomic) particles: electrons, protons and neutrons. (Hydrogen atom lacks neutrons). ⚛️
    - (ii) The nucleus is located at the centre of the atom. It contains protons and neutrons, which account for the total mass of that atom. 🎯
    - (iii) The nucleus is positively charged due to the presence of protons in it. ➕
    - (iv) The electrons are outside the nucleus and have negligible mass. ➖
    - (v) The number of electrons in an atom is equal to the number of protons in it, hence the atom is electrically neutral. ⚖️
    - (vi) The electrons revolve rapidly in fixed circular paths around the nucleus. These circular paths are called energy levels or shells or orbits. 💫
    - (vii) The atoms of different elements contain different numbers of electrons, protons and neutrons. 🔢
- **Structure of an Atom (Summary):**
    - An atom is built up of a number of sub-atomic particles. The three sub-atomic particles of great importance in understanding the structure of an atom are electrons, protons and neutrons. 🧱

- There are two structural parts of an atom, the nucleus and the empty space in which there are imaginary paths called orbits. 🌌

- **Bohr-Bury Scheme for Distribution of Electrons in Orbits:**
  - **Rule 1:** The maximum possible number of electrons in a particular shell is given by the formula $2n^2$ where n denotes the serial number of that shell.
    - First or K shell has $(2 \times 1^2) = 2$ electrons
    - Second or L shell has $(2 \times 2^2) = 8$ electrons
    - Third or M shell has $(2 \times 3^2) = 18$ electrons
    - Fourth or N shell has $(2 \times 4^2) = 32$ electrons, and so on. 🔢
  - **Rule 2:** Electrons are not accommodated in a given shell, unless the inner shells are filled. In other words the shells are filled in stepwise manner. 🪜
  - **Rule 3:** The outermost shell of an atom cannot accommodate more than 8 electrons, even if it has the capacity to accommodate more electrons. 🛡️
  - **Rule 4:** The outermost shell of a chemically stable atom can accommodate a maximum of 8 electrons, except Hydrogen and Helium atoms which have only one shell, and can thus accommodate a maximum of 2 electrons only. 🎈

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Symbolic representation of Element:**
  - ${}^{A}_{Z}X$
  - X denotes the symbol of the element.
  - Superscript A = Mass number.
  - Subscript Z = Atomic number.
  - *Example:* ${}^{35}_{17}Cl$ (Atomic number $[Z] = p = e = 17$; Mass number $[A] = p + n = 35$; No. of neutrons $[n] = A - Z = 18$). ⚛️

- **Formulas:**

- Atomic number $[Z]$ = no. of protons $[p]$ = no. of electrons $[e]$ ($Z = p = e$)
- Mass number $[A]$ = no. of neutrons $[n]$ + no. of protons $[p]$ ($A = n + p$)
- No. of neutrons $[n] = A - Z$

- **Properties of sub-atomic particles (Table 4.2):**

Particle	Symbol	Charge (electronic unit)	Mass (amu)	Mass (grams)
1. Electron	${}_{-1}^{0}e$ or e^-	-1	$1/1840$	9.1×10^{-28} g
2. Proton	${}_{1}^{1}H$ or p^+	$+1$	1	1.6×10^{-24} g
3. Neutron	${}_{0}^{1}n$ or n	0	1	1.6×10^{-24} g

📖 1. Core Definitions & Key Terminology 📝

- **Valence Shell:** The outermost shell of an atom. 🛡️
- **Valence Electrons:** The electrons present in the valence shell. 💫
- **Cations:** Elements that can lose electrons to form ions, which are positively charged (e.g., metals). ➕
- **Anions:** Elements that can gain electrons to form ions, which are negatively charged (e.g., non-metals). ➖
- **Ion:** An ion is an electrically charged atom or group of atoms formed by the loss or gain of electrons. ⚡
- **Inert Gases (Noble or Rare Gases):** Elements with the valence shell having 8 electrons, and helium with 2 electrons, cannot gain or lose electrons so they are chemically inactive. 🛑
- **Octet:** An arrangement of eight electrons in the outermost shell. 🎱

- **Octet Rule:** Elements tend to combine with one another to attain the stable electronic configuration of the nearest inert gas, i.e. 8 electrons in its valence shell. This is known as the octet rule. 🎯
- **Duplet Rule:** In the case of helium, the valence shell has 2 electrons, which represents a stable configuration known as the duplet rule. ✌️
- **Valency:** The number of electrons gained, lost or shared to attain the octet in the outermost shell gives the combining capacity of the element, that is, its valency. 🤝
- **Chemical Bond:** As a result of the redistribution of electrons, a force of attraction develops between atoms, which binds them together to form molecules. This force is known as a chemical bond. 🔗
- **Isotopes:** Atoms of the same element having the same atomic number but different mass numbers. ⚖️
- **Isotopy:** The phenomenon of an element showing similar chemical properties but possessing different mass is said to represent ISOTOPY. ⚖️

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why do elements with the same number of valence electrons show similar properties?** The chemical properties of elements are decided by their valence electrons, since they are the ones that take part in chemical reactions. Therefore, elements with the same number of valence electrons exhibit similar properties, while those with different numbers show different chemical properties. 🧪
- **How does the size of ions compare to their parent atoms?**
    - The size of a cation is less than that of the atom. 📉
    - The size of an anion is greater than that of the atom. 📈
- **What happens when an atom changes to an ion?**
    - The number of protons remains the same.
    - The number of neutrons remains the same.
    - The number of electrons changes. ⚡

- **Why are noble gases unreactive?** Atoms of all noble gases (except Helium) have eight electrons in their outermost shell (an octet). Helium has 2 electrons (duplet) in its only shell, which can occupy a maximum of 2 electrons (according to the $2n^2$ rule). So it is a stable element, even though the 8-electron rule does not apply in its case. Since their outer shells are already filled, they do not combine either with other atoms or among themselves to form molecules; their combining capacity or valency is zero. 🛑
- **What determines the chemical activity of an atom?** The chemical activity of an atom depends upon the number of electrons in the valence shell of its atom. Chemically active atoms have an incomplete octet (less than 8 electrons) or less than 2 electrons (in the case of hydrogen) in their valence shell. ⚙️
- **Why do atoms combine? (Kossel and Lewis, 1916)** The atoms of all elements, other than the inert gases, combine because they have incomplete valence shells and tend to attain a stable configuration (electronic theory of valency). This is achieved either by gaining electrons, losing electrons, or sharing electrons with other atoms. 🤝
- **Why do isotopes have identical chemical properties?** Isotopes have the same atomic number (Z), i.e., the same number of electrons and protons, and hence the same electronic configuration. Chemical properties are determined by the electronic configuration of an atom. Thus, isotopes of an element are chemically alike. 🧪
- **Why do isotopes differ in mass?** It is noticed that two or more atoms of the same element may contain the same number of protons but they may differ in the number of neutrons. The varying number of neutrons causes the difference in mass numbers. ⚖️
- ✍️ *Handwritten Margin Note:* "OIL RIG" (Written under the anions section, a common mnemonic: Oxidation Is Loss, Reduction Is Gain). 📝
- ✍️ *Handwritten Margin Note:* "loss of e- → oxidation" (Written next to the cation section). 📝
- ✍️ *Handwritten Margin Note:* "to tough" (Written with a crossed-out star next to the table of Inert Gas Configurations). 📝

- ✍️ *Handwritten Margin Note:* "neutrons (mass diff)" (Written pointing to the heading 'Isotopes'). 📝
- ✍️ *Handwritten Margin Note:* "same e^-" (Written next to the definition of isotopes). 📝
- ✍️ *Handwritten Margin Note:* "learn" (Written next to the 'Examples of Isotopes' table). 📝
- ✍️ *Handwritten Margin Note:* "diff $-Z$" (Written next to the 'Examples of Isotopes' table, likely referring to $A - Z$ for finding neutrons). 📝
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📌 3. Rules, Principles & Characteristics ⚙️

- **Valency Rules based on Valence Electrons:**
  - **Metals:** Elements which have valence electrons 1 or 2 or 3 (except hydrogen and helium) are metals. They can lose electrons to form cations. Metals are good reducing agents. 🥇
  - **Non-metals:** Elements with $4,5,6,$ or 7 electrons in their valence shells are non-metals. They can gain electrons to form anions. Non-metals are good oxidising agents. 🪵
  - **Calculating Valency:** Valency of the elements having $1,2,$ or 3 valence electrons is $1,2,$ or 3 respectively, while the valency of elements with $4,5,6,$ and 7 valence electrons is $4,3,2,$ and 1 (8 minus valence electrons) respectively. 🧮
- **Formation of Sodium Ion (Na^+):**
  - $Na(2,8,1) - e^- \rightarrow Na^+(2,8)$ ⚡
- **Formation of Chlorine Ion (Cl^-):**
  - $Cl(2,8,7) + e^- \rightarrow Cl^-(2,8,8)$ ⚡
- **Properties of Isotopes:**
  - Isotopes have the same atomic number (Z).
  - Isotopes have different mass numbers (A).

- Isotopes have similar chemical properties (due to identical electron configuration).
- Isotopes differ only in their number of neutrons. ⚖️

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Table 4.4: Electronic configuration of the inert or rare gases**

Inert gas	Atomic No.	K	L	M	N	O	P
He	2	2					
Ne	10	2	8				
Ar	18	2	8	8			
Kr	36	2	8	18	8		
Xe	54	2	8	18	18	8	
Rn	86	2	8	18	32	18	8

- **Examples of Isotopes (from table and text):**

Element	Number of isotopes	Symbols
1. Hydrogen	Three	${}_{1}^{1}H, {}_{1}^{2}H, {}_{1}^{3}H$
2. Carbon	Three	${}_{6}^{12}C, {}_{6}^{13}C, {}_{6}^{14}C$
3. Chlorine	Two	${}_{17}^{35}Cl, {}_{17}^{37}Cl$
4. Oxygen	Two	${}_{8}^{16}O, {}_{8}^{18}O$
5. Potassium	Two	${}_{19}^{39}K, {}_{19}^{40}K$

- **Detailed Breakdown of Specific Isotopes:**
    - **Hydrogen ($Z = 1$):**
        - Protium ($_1^1H$): Mass 1, Neutrons 0, Protons 1, Electrons 1.
        - Deuterium ($_1^2H$): Mass 2, Neutrons 1, Protons 1, Electrons 1. (Also known as heavy hydrogen).
        - Tritium ($_1^3H$): Mass 3, Neutrons 2, Protons 1, Electrons 1.
    - **Carbon ($Z = 6$):**
        - Carbon-12 ($_6^{12}C$): Mass 12, Neutrons 6, Protons 6, Electrons 6.
        - Carbon-13 ($_6^{13}C$): Mass 13, Neutrons 7, Protons 6, Electrons 6.
        - Carbon-14 ($_6^{14}C$): Mass 14, Neutrons 8, Protons 6, Electrons 6.
    - **Chlorine ($Z = 17$):**
        - Chlorine-35 ($_{17}^{35}Cl$): Mass 35, Neutrons 18 $(35 - 17)$, Protons 17, Electrons 17.
        - Chlorine-37 ($_{17}^{37}Cl$): Mass 37, Neutrons 20 $(37 - 17)$, Protons 17, Electrons 17.

📖 1. Core Definitions & Key Terminology 📝

- **Isobars:** Atoms of different elements with the same mass number, but different atomic numbers. ⚖️
- **Radioactive Isotopes (Radioisotopes):** Isotopes like tritium and carbon-14 that are unstable so their nuclei break up spontaneously emitting certain types of radiation. ☢️
- **Radioactivity:** The spontaneous decay of unstable radioisotopes. 💥
- **Electrovalent (or Ionic) Bond:** The chemical bond formed as a result of the transfer of electrons from one atom of an element to one atom of another element. 🤝⚡
- **Electrovalent Compound:** The compound formed by the transfer of electrons. 🧂

- **Electrovalency:** The number of electrons that an atom of an element loses or gains to form an electrovalent bond. 🔢
- **Ion:** A charged particle which is formed due to the gain or loss of one or more electrons by an atom. ⚡
- **Electropositive Element (Cation):** A metallic element whose one atom readily loses electron(s) to form a positively charged ion. ➕
- **Electronegative Element (Anion):** A non-metallic element whose atom readily accepts electron(s) to form a negatively charged ion. ➖
- **Electron Dot Symbol (Lewis Symbol):** A symbol consisting of the symbol of the element surrounded by dots representing the outermost shell electrons (valence electrons). ⚛️

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why do isotopes have different physical properties?** Physical constants or physical properties depend on the atomic mass (mass number). Since isotopes have different mass numbers (due to different numbers of neutrons), their physical properties (like density, boiling point, etc.) differ. For example, the boiling point of protium (^1H) is 20.38 K and that of deuterium (^2H) is 23.5 K. The higher the mass of the isotope of an element, the higher is the boiling point. 🌡️
- **Why do isotopes of an element undergo the same chemical reactions?** Chemical properties depend on the atomic number (number of electrons and protons), which is the same for all isotopes of an element. Therefore, they have similar chemical properties (e.g., all isotopes of chlorine undergo the same reactions). 🧪
- **Why do elements have fractional atomic masses?** An atom can have a fractional atomic mass because the atomic mass of an element represents the weighted average mass of all the naturally occurring isotopes of that element. For example, natural chlorine contains 75% of ^{35}Cl and 25% of ^{37}Cl (ratio $3:1$). Average mass = $\frac{35\times3+37\times1}{4} = 35.5$. ⚖️
- **Why are Isobars chemically different?** Since the properties of elements depend upon atomic number, and isobars have different atomic numbers, they have both different physical and chemical properties. ❌

- **Why are ionic compounds stable?** While ionic compounds are formed by ions and there exists a repulsive force between ions of like charges, the electrostatic force of attraction between opposite charges is much higher. It makes the ionic compounds stable. 🧲
- **Why do metals and non-metals form ionic bonds?** Metals have one, two, or three electrons in their valence shell and have a tendency to lose them. Non-metals have five, six, or seven electrons in their valence shell and have a tendency to gain electrons. Thus, they complement each other to form ionic bonds. 🤝
- **Why did Lewis introduce dot diagrams?** To show bond formation by orbit diagram is not so convenient as electrons of only outermost orbits are taking part in bond formation. Lewis gave a concept of dot diagram in which only valence electrons are shown to facilitate study. 🎯
- **How to distinguish electrons from different atoms in Lewis structures?** Symbols other than dots, such as circles and crosses can be used. For example, if dots represent the outermost electrons of nitrogen, and crosses represent the electrons of hydrogen, ammonia (NH_3) can be clearly represented. ❌🔴
- ✍️ *Handwritten Margin Note:* "Mass ↑ B.P. ↑" (Written next to the explanation of boiling points of isotopes). 📝
- ✍️ *Handwritten Margin Note:* "Same mass, diff Z" (Written next to the definition of Isobars). 📝
- ✍️ *Handwritten Margin Note:* "Carbon dating" (Written next to the use of Carbon-14). 📝
- ✍️ *Handwritten Margin Note:* "Imp" (Written next to the Isobar examples). 📝
- ✍️ *Handwritten Margin Note:* "imp" (Written next to the definition of Electrovalent bond). 📝
- ✍️ *Handwritten Margin Note:* "Na → Cs" (Written showing the increasing tendency to lose an electron down a group, highlighting Cesium as maximum). 📝

- ✍️ *Handwritten Margin Note:* "$F \rightarrow I$" (Written showing halogens, highlighting Fluorine has maximum tendency to gain an electron). 📝
- ✍️ *Handwritten Margin Note:* "Verry imp" (Written next to the electron dot structure of $NaCl$ and $MgCl_2$). 📝
- 🧩 *Page-Shift Stitch:* "Bonds formed between metals and non-metals are ionic or electrovalent bonds." (Seamlessly connects the concept from the bottom of page 92 to the top of page 93). 🔗

📌 3. Rules, Principles & Characteristics ⚙️

- **Uses of Isotopes:**
    1. Isotope of cobalt ^{60}Co is used in radiotherapy for treating cancer and other diseases. ⚕️
    2. ^{14}C is used for determining the age of historical and geological material (Carbon dating). ⏳
    3. An isotope of Uranium ^{235}U is used as a fuel in nuclear reactors. ☢️
    4. An isotope of Iodine ^{131}I is used in the treatment of goitre. 💊
    5. Radioactive isotopes are used in industry to detect the leakage in underground oil pipelines, gas pipelines and water pipes. 🛢️
- **Features common to ionic bonding:**
    - Metal atoms always lose their outer electrons to form positive ions. ➕
    - Non-metal atoms with one exception of hydrogen, always gain electrons to become negative ions. ➖
    - In both cases the ions formed have a more stable electron arrangement, that of the noble gas nearest to the element concerned. 🎯
    - Ionic bonds result from the attraction between oppositely charged ions (Coulomb forces). 🧲
- **Electron Dot Symbol Rules:**
    - The symbol of the element is written in the centre.

- Dots (or crosses) are placed around it representing ONLY the valence electrons.
- Paired electrons are represented by a pair of dots.
- Unpaired electrons are represented by a single dot. 🔴
- Transfer of an electron is shown by a curved arrow. ↪️

📊 4. Master Data, Configurations, Formulas & Tables 🍼

- **Examples of Isobars:**
  - (i) ${}^{40}_{18}Ar$ and ${}^{40}_{20}Ca$ (Argon and Calcium: same mass 40, different Z). ⚖️
  - (ii) ${}^{54}_{24}Cr$ and ${}^{54}_{26}Fe$ (Chromium and Iron: same mass 54, different Z). ⚖️
- **Examples of Electrovalent (Ionic) Compounds:**
  - $NaCl$ (Sodium chloride)
  - $MgCl_2$ (Magnesium chloride)
  - CaO (Calcium oxide)
  - KBr (Potassium Bromide)
  - $CaCl_2$ (Calcium chloride) 🧂
- **Formation of Sodium Chloride ($NaCl$):**
  - Na **Atom:** Atomic No. 11, Config: 2,8,1. Tends to lose $1\ e^-$.
    - $Na(2,8,1) \rightarrow Na^+(2,8) + 1e^-$
  - Cl **Atom:** Atomic No. 17, Config: 2,8,7. Tends to gain $1\ e^-$.
    - $Cl(2,8,7) + 1e^- \rightarrow Cl^-(2,8,8)$
  - **Bonding:** The cation Na^+ and anion Cl^- are attracted towards each other due to opposite electrical charges (coulomb force) forming an ionic compound. ⚡
  - *Electron Dot Structure:* $Na \cdot + \cdot \ddot{Cl}: \rightarrow [Na]^+[\cdot \ddot{Cl}:]^-$
- **Formation of Magnesium Chloride ($MgCl_2$):**

- Mg **Atom:** Atomic No. 12, Config: $2,8,2$. Tends to lose $2\ e^-$.
  - $Mg(2,8,2) \rightarrow Mg^{2+}(2,8) + 2e^-$
- Cl **Atom:** Atomic No. 17, Config: $2,8,7$. Can only accept $1\ e^-$.
- **Ratio:** Therefore, for each magnesium atom forming a magnesium ion, there must be two chlorine atoms to form two chloride ions. Ratio of Mg to Cl is $1:2$.
  - $2Cl(2,8,7) + 2e^- \rightarrow 2Cl^-(2,8,8)$
- *Electron Dot Structure:* $:\ddot{Cl}\cdot + \cdot Mg \cdot + \cdot \ddot{Cl}: \rightarrow [Mg]^{2+}2\left[\cdot \ddot{Cl}:\right]^-$

- **Formation of Calcium Oxide (CaO) - Initial setup:**
  - Ca **Atom:** Atomic No. 20, Config: $2,8,8,2$. Tends to lose $2\ e^-$ to form Ca^{2+} ion.
  - O **Atom:** Atomic No. 8, Config: $2,6$. Tends to gain $2\ e^-$ to form O^{2-} ion.

📖 1. Core Definitions & Key Terminology 📝

- **REDOX REACTION:** A reaction where the transfer of electron(s) is involved. The electropositive atom loses electrons and undergoes oxidation, while the electronegative atom gains electrons and undergoes reduction. 🔄
- **Oxidising Agent:** An acceptor of electron(s). 🧲
- **Reducing Agent:** A donor of electron(s). 🎁
- **Covalent (Molecular) Bond:** The chemical bond that is formed between two combining atoms by mutual sharing of one or more pairs of electrons. 🤝
- **Covalent Compound / Molecule:** The compound or molecule formed due to the sharing of electrons (covalent bond). 🔗
- **Single Covalent Bond:** Formed by the sharing of *one pair* of electrons between the atoms, each contributing one electron. Denoted by a short line (—). ➖

- **Double Covalent Bond:** Formed by the sharing of *two pairs* of electrons between two atoms. It is a combination of two single bonds, represented by two short lines (=). 🟰
- **Triple Covalent Bond:** Formed by the sharing of *three pairs* of electrons between two atoms. It is a combination of three single bonds, represented by three short lines (≡). ☰
- **Covalency:** The number of an atom's electrons taking part in the formation of shared pairs. 🔢
- **Non-polar Covalent Compounds:** Covalent compounds where the shared pair of electron(s) are equally distributed between the two atoms. No charge separation takes place. The molecule is symmetrical and electrically neutral. ⚖️
- **Polar Covalent Compounds:** Covalent compounds where the shared pair of electrons are not at an equal distance between the two atoms. This results in the development of fractional positive and negative charges on them, and they ionize in water. ⚡

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why is the formula for calcium oxide CaO and not Ca_2O?** Since only *one* oxygen atom (needs 2 electrons) is needed to accept the 2 valence electrons donated by a single calcium atom, the resulting formula is CaO. ⚖️
- **Why do oxidation and reduction always occur simultaneously?** Because the electron(s) lost by the reducing agent must be exactly gained by the oxidising agent. They cannot happen in isolation. 🔄
- **Why do non-metals form covalent bonds instead of ionic ones?** The atoms of non-metals usually have $5,6$, or 7 electrons in their outermost shell (except Carbon with 4, Hydrogen with 1). These elements do not favour the loss of electrons due to *energy considerations*, making electron transfer impossible. Therefore, they can complete their octet only by mutually sharing electrons. 🛡️
- **Why are perfectly identical non-metal molecules non-polar?** If the two covalently bonded atoms are identical (like H_2, Cl_2, O_2), the shared

electron pair(s) is at an equal distance from the combining atoms. It is equally attracted by the identical nuclei of both atoms, preventing any charge separation. 🧲

- ✍️ *Handwritten Margin Note:* "OIL RIG" (Written next to the definition of oxidising/reducing agents to remember: Oxidation Is Loss, Reduction Is Gain of electrons). 📝
- ✍️ *Handwritten Margin Note:* "$2Na + Cl_2 \rightarrow 2NaCl$" with "oxid" written under Na and "red" written under Cl. 📝
- ✍️ *Handwritten Margin Note:* "FON" (Written above the polar covalent bond section, likely a teacher's tip referring to highly electronegative elements Fluorine, Oxygen, Nitrogen that cause polarity). 📝
- ✍️ *Handwritten Margin Note:* "imp" (Written next to the physical properties of electrovalent compounds). 📝
- 🧩 *Page-Shift Stitch:* "Similarly, a double bond is formed by the sharing of two pairs of electrons between two atoms. A double bond is actually a combination of two single bonds, so it is represented by putting two short lines (=) between the two atoms." (Successfully stitched from the bottom of page 96 to the top of page 97). 📝

📌 3. Rules, Principles & Characteristics ⚙️

- **Properties of Electrovalent (Ionic) Compounds:**
    - *State:* They are hard, brittle, crystalline solids consisting of ions, held together by strong electrostatic forces of attraction which cannot be separated easily. 🧊
    - *Melting/Boiling Points:* They are non-volatile, with high boiling and high melting points. 🔥
    - *Conductivity:* They do **not** conduct electricity in the solid state, but they are **good conductors** of electricity in the fused (melted) or aqueous (dissolved in water) state. ⚡
- **Rules for Covalent Bonding:**
    - Each combining atom contributes an *equal* number of electron(s) for sharing. ⚖️

- A covalent bond is formed when both combining atoms have four or more electrons in their outermost shells (i.e., non-metals). *Exceptions are* H, Be, B, *etc.* 🛑
- Hydrogen can combine with all non-metals with the help of covalent bonds. 🎈
- A covalent bond can also be formed between two atoms of the *same* non-metal (e.g., Cl_2). 👬

- **Complex Molecule Bonding Combinations:**
  - *Ethene (Ethylene,* C_2H_4*):* Has a combination of one double covalent bond ($C = C$) and four single covalent bonds ($C - H$). 🔗
  - *Ethyne (Acetylene,* C_2H_2*):* Has one triple covalent bond ($C \equiv C$) and two single covalent bonds ($C - H$). ☰
- **Standard Covalencies:**
  - Hydrogen = 1
  - Oxygen = 2
  - Nitrogen = 3
  - Carbon = 4 🔢

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Redox Half-Reactions (Formation of** $NaCl$**):**
  - **Oxidation (Loss of** e^-**):** $2Na \rightarrow 2Na^+ + 2e^-$
  - **Reduction (Gain of** e^-**):** $Cl_2 + 2e^- \rightarrow 2Cl^-$
  - **Net Redox Reaction:** $2Na + Cl_2 \rightarrow 2Na^+ + 2Cl^-$ (or $2NaCl$) 🧂
- **Redox Identification in Displacement Reaction:**
  - **Reaction:** $Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$
  - **Ionic breakdown:** $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu(s)$
  - **Oxidation:** $Zn - 2e^- \rightarrow Zn^{2+}$ (Zn is the Reducing Agent).
  - **Reduction:** $Cu^{2+} + 2e^- \rightarrow Cu$ (Cu^{2+} is the Oxidising Agent). ⚡

- **Step-by-Step Formation of Non-Polar Covalent Molecules:**
    - **1. Hydrogen Molecule (H_2):**
        - *Config:* H has 1 valence electron. Needs 1 more for a duplet.
        - *Mechanism:* Two H atoms share 1 pair of electrons.
        - *Result:* Single covalent bond ($H-H$). 🎈
    - **2. Chlorine Molecule (Cl_2):**
        - *Config:* Cl $(2,8,7)$. Nearest noble gas Argon $(2,8,8)$. Needs 1 electron for octet.
        - *Mechanism:* Two Cl atoms contribute 1 electron each. They have 1 shared pair.
        - *Result:* Single covalent bond ($Cl-Cl$). 🧪
    - **3. Oxygen Molecule (O_2):**
        - *Config:* O $(2,6)$. Nearest noble gas Neon $(2,8)$. Needs 2 electrons for octet.
        - *Mechanism:* Two O atoms contribute 2 electrons each. They have 2 shared pairs.
        - *Result:* Double covalent bond ($O=O$). 🫁
    - **4. Nitrogen Molecule (N_2):**
        - *Config:* N $(2,5)$. Nearest noble gas Neon $(2,8)$. Needs 3 electrons for octet.
        - *Mechanism:* Two N atoms contribute 3 electrons each. They have 3 shared pairs.
        - *Result:* Triple covalent bond ($N\equiv N$). ⚪

📊 4. Master Data, Configurations, Formulas & Tables (ADDENDUM) 🧪

- **Completion of Calcium Oxide (CaO) Formation (From Page 95):**
    - *Equation:* $Ca-2e^- \rightarrow Ca^{2+}$ (Cation config: $2,8,8$)
    - *Equation:* $O+2e^- \rightarrow O^{2-}$ (Anion config: $2,8$)

- *Final Electron Dot Structure:* $Ca: + \cdot \ddot{O}: \rightarrow Ca^{2+}[:\ddot{O}:]^{2-}$ 🔗

- **Specific Examples of Single Covalent Bonds (From Page 96):**
  - *Chlorine:* $Cl - Cl$
  - *Hydrogen chloride:* $H - Cl$
  - *Ammonia:* $H - N - H$ (with one H below the N)
  - *Water:* $H - O - H$
  - *Methane:* C surrounded by four single bonds to H atoms.
  - *Carbon tetrachloride:* C surrounded by four single bonds to Cl atoms. 🔗

- **Specific Examples of Multiple Bonds (From Page 97):**
  - **Ethene (Ethylene, C_2H_4):** Electron dot diagram shows carbon atoms sharing *two pairs* of electrons ($C = C$) and each carbon sharing *one pair* with two hydrogens.
  - **Ethyne (Acetylene, C_2H_2):** Electron dot diagram shows carbon atoms sharing *three pairs* of electrons ($C \equiv C$) and each carbon sharing *one pair* with one hydrogen. ☰

📖 1. Core Definitions & Key Terminology 📝

- **Hydrogen chloride molecule:** A polar covalent compound formed by the sharing of one electron pair between a hydrogen atom and a chlorine atom. 🔗
- **Water molecule:** A polar covalent compound formed by the sharing of two electron pairs, one pair between the oxygen atom and each of the two hydrogen atoms. 💧
- **Ammonia molecule:** A polar covalent compound formed by the sharing of three electron pairs, one pair between the nitrogen atom and each of the three hydrogen atoms. ☁️
- **Carbon tetrachloride molecule:** A non-polar covalent compound formed by the sharing of four electron pairs, one pair between the carbon atom and each of the four chlorine atoms. 🧪

- **Methane molecule:** A non-polar covalent compound formed by the sharing of four electron pairs, one pair between the carbon atom and each of the four hydrogen atoms. 🫧
- **Homolytic bond fission (or cleavage):** The symmetrical breaking of a covalent bond where each atom retains one electron from the original shared pair, resulting in two neutral species with unpaired electrons. 🪓
- **Free radicals:** Neutral species containing unpaired electrons, formed by homolytic bond cleavage. They are unstable, short-lived, and highly reactive. ⚡
- **Photolysis:** The process of homolytic bond cleavage that requires energy to overcome bond dissociation energy, typically supplied by heat or high-energy radiation such as UV light. 🌞

💡 2. The "Why", Scientific Reasons & Extra Facts 🧠

- **Why does hydrogen chloride form a single covalent bond?** To attain the stable electronic configuration of the nearest noble gas, hydrogen needs one electron (to reach helium's duplet), and chlorine also needs one electron (to reach argon's octet). By sharing one electron pair, both atoms achieve stability. 🤝
- **Why are covalent compounds generally gases, liquids, or soft solids?** Covalent compounds are made up of molecules held together by weak forces of attraction (intermolecular forces), unlike the strong electrostatic forces in ionic lattices. ☁️
- **Why do covalent compounds have low melting and boiling points?** Because they are held together by weak intermolecular forces, minimal thermal energy is required to separate the molecules, making them volatile with low melting and boiling points. 📉
- **Why are covalent compounds non-conductors of electricity?** They do not contain free mobile ions or electrons in their solid, molten, or aqueous states (if non-polar). 🔌🚫
- **Why do *polar* covalent compounds sometimes conduct electricity?** While non-polar compounds do not ionize, some covalent compounds are polar in nature. These polar molecules can form ions and ionize in their

aqueous solutions, allowing them to act as electrolytes and conduct current. ⚡ 💧

- **Why do free radicals form?** They form when energy (like UV light) overcomes the bond dissociation energy, causing a covalent bond to break symmetrically (homolytically). 🌞 🪓
- **Why are free radicals highly reactive?** The presence of an unpaired electron makes the species inherently unstable, driving it to rapidly react to pair the electron and achieve stability. ⚡
- **Historical Context (Chapter at a Glance):** The idea of the smallest unit of matter i.e. PARMANU which combines to give ANU was given by the Indian scientist Maharshi Kanada. Greek philosopher Democritus called the parmanu an 'ATOM'. 🧘 🏛️
- ✍️ *Handwritten Margin Note:* "Note cross is imp" (Written next to the electron dot structure of Hydrogen chloride, emphasizing the use of crosses and dots to distinguish electrons). 📝
- ✍️ *Handwritten Margin Note:* "other element" (Written next to a circled 'X' symbol in the water molecule diagram, indicating that crosses represent electrons from the other element, hydrogen). 📝
- ✍️ *Handwritten Margin Note:* "of carbon" (Written with an arrow pointing to the 'x' in the carbon tetrachloride electron dot diagram, clarifying that the cross represents carbon's valence electron). 📝
- ✍️ *Handwritten Margin Note:* "Verry imp" (Written pointing to the "Note" box summarizing the physical properties of covalent compounds). 📝
- ✍️ *Handwritten Margin Note:* Hand-drawn mechanism showing $Cl - Cl$ bond breaking symmetrically with half-arrows pointing to each Cl atom, resulting in two $Cl\cdot$ radicals. 📝
- 🧩 *Page-Shift Stitch:* "These compounds are insoluble in water but dissolve in organic solvents." (Successfully stitched from the bottom of page 100 to the top of page 101, completing the physical properties of covalent compounds). 📝

📌 3. Rules, Principles & Characteristics ⚙️

- **Properties of Covalent Compounds (Summary):**
    - *Physical State:* They are usually gases, liquids, or soft solids. ☁️
    - *Melting/Boiling Points:* They are volatile, with low boiling and low melting points. 📉
    - *Conductivity:* They are non-conductors of electricity in solid, molten, or aqueous states (unless they are polar and ionize in water). 🔌 🚫
    - *Solubility:* These compounds are insoluble in water but dissolve in organic solvents. 💧 🚫
- **Homolytic Cleavage Principle:** During homolytic bond fission, each atom retains exactly one electron from the original shared covalent bond. ⚖️

📊 4. Master Data, Configurations, Formulas & Tables 🧪

- **Formation of Hydrogen Chloride (HCl):**
    - *Atoms:* Hydrogen ($H, 1$, needs 1), Chlorine ($Cl, 2{,}8{,}7$, needs 1).
    - *Structure:* One shared pair of electrons ($H \times\cdot \ddot{Cl}:$ or $H - Cl$).
    - *Total Atoms:* 2 ($1H, 1Cl$).
- **Formation of Water (H_2O):**
    - *Atoms:* Hydrogen ($H, 1$, needs 1), Oxygen ($O, 2{,}6$, needs 2).
    - *Structure:* Two shared pairs of electrons ($H \times\cdot \ddot{O} \cdot\times H$ or $H - O - H$).
    - *Total Atoms:* 3 ($2H, 1O$).
- **Formation of Ammonia (NH_3):**
    - *Atoms:* Nitrogen ($N, 2{,}5$, needs 3), Hydrogen ($H, 1$, needs 1).
    - *Structure:* Three shared pairs of electrons (Central N with lone pair, surrounded by $3H$ atoms: $H \times\cdot \ddot{N} \cdot\times H$, with one H below).
    - *Total Atoms:* 4 ($1N, 3H$).
- **Formation of Carbon tetrachloride (CCl_4):**

- *Atoms:* Carbon ($C, 2{,}4$, needs 4), Chlorine ($Cl, 2{,}8{,}7$, needs 1).
- *Structure:* Four shared pairs of electrons (Central C sharing one pair with 4 surrounding Cl atoms).
- *Total Atoms:* 5 ($1C, 4Cl$).

- **Formation of Methane (CH_4):**
  - *Atoms:* Carbon ($C, 2{,}4$, needs 4), Hydrogen ($H, 1$, needs 1).
  - *Structure:* Four shared pairs of electrons (Central C sharing one pair with 4 surrounding H atoms).
  - *Total Atoms:* 5 ($1C, 4H$).

- **Equation for Homolytic Bond Cleavage (Photolysis of Chlorine):**
  - $:\ddot{Cl}:\ddot{Cl}: \xrightarrow{UV} :\ddot{Cl}\cdot + \cdot\ddot{Cl}:$
  - $Cl_2 \rightarrow Cl^{\cdot} + Cl^{\cdot}$ (Chlorine molecule → Free radical + Free radical).